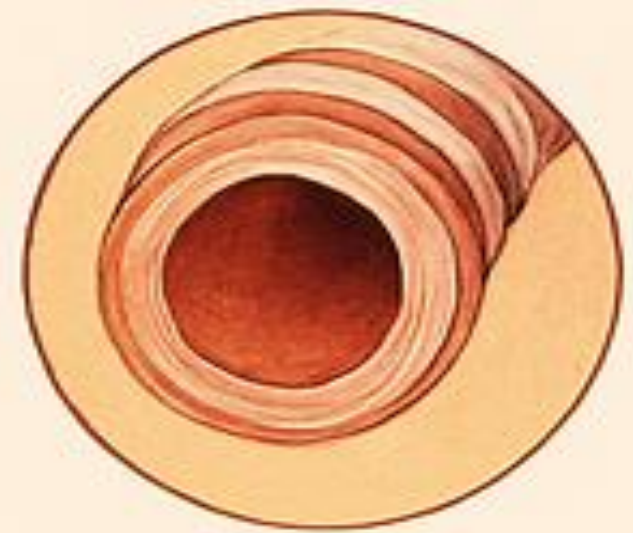


ASTHMA

DR/ ZEINAB



Normal
bronchus



Asthmatic
bronchus

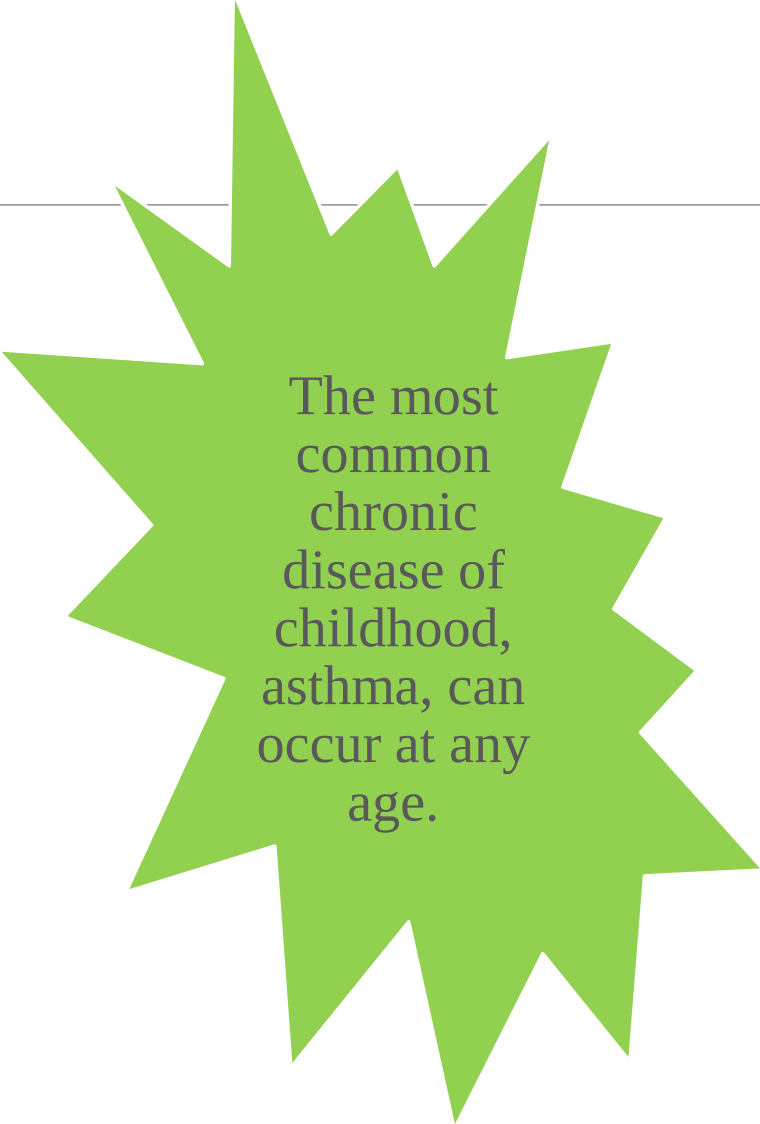


AGENDA

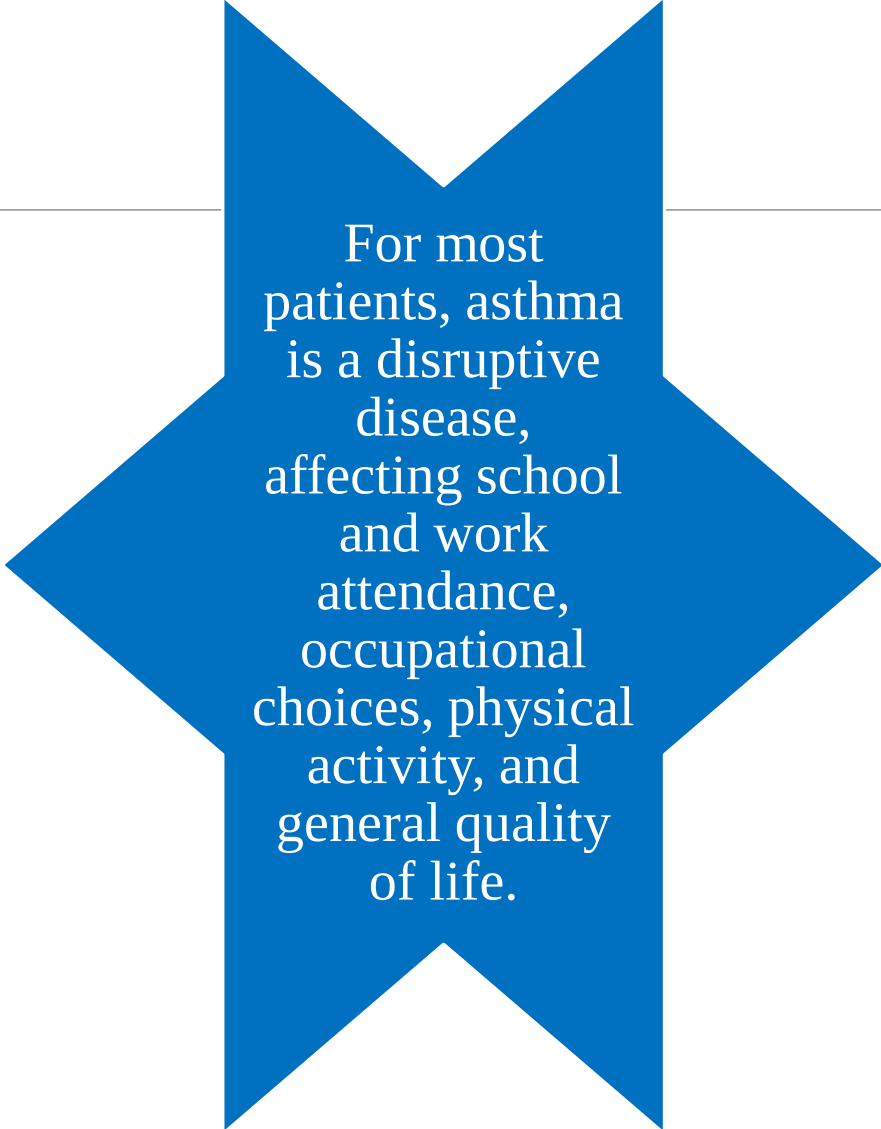
- Definition of bronchial asthma
- Causes of bronchial asthma
- Pathophysiology of bronchial asthma
- Clinical Manifestations of bronchial asthma
- Assessment and diagnostic findings of bronchial asthma
- Prevention of bronchial asthma
- Complications of bronchial asthma
- Medical & nursing management of bronchial asthma



- **Asthma** is a chronic inflammatory disease of the airways that causes airway hyper-responsiveness, mucosal edema, and mucus production.

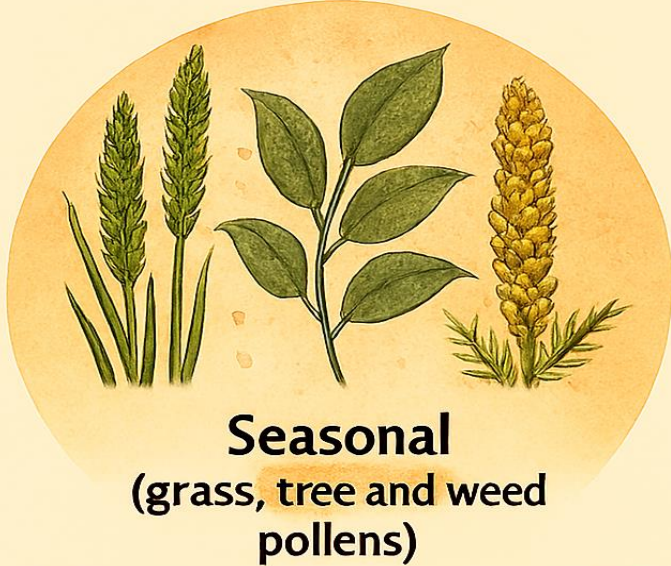


The most
common
chronic
disease of
childhood,
asthma, can
occur at any
age.



For most
patients, asthma
is a disruptive
disease,
affecting school
and work
attendance,
occupational
choices, physical
activity, and
general quality
of life.

COMMON ALLERGENS



Perennial
(eg.-dust,
animal dander)



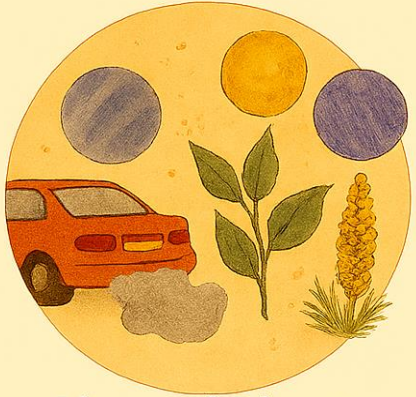
Perennial
(eg.-dust,
animal dander)

ENGAGING THE AUDIENCE

- Allergy is the strongest predisposing factor for asthma.
- Chronic exposure to airway irritants or allergens also increases the risk of asthma.
- Common allergens can be seasonal (grass, tree, and weed pollens) or perennial (eg, dust, animal dander).

COMMON TRIGGERS

Common triggers for asthma symptoms and exacerbations include



Airway irritants

(air pollutants, cold, heat, weather changes, strong odors or perfumes, smoke)



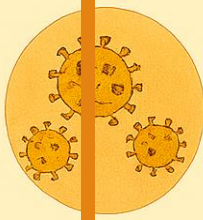
Exercise



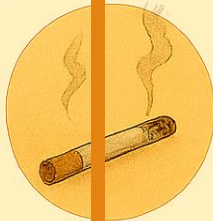
Stress or emotional upset



Stress or emotional upset



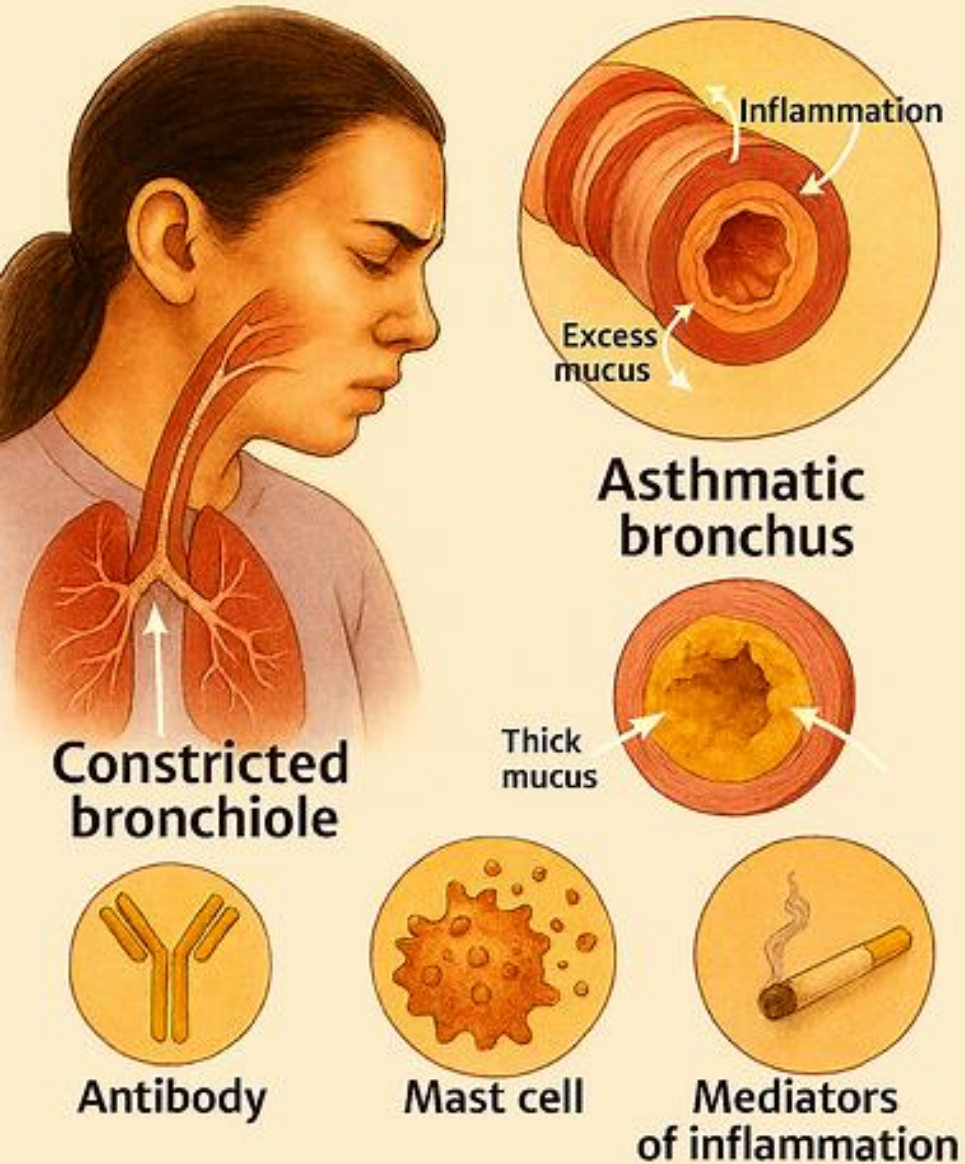
Dust



Common triggers for asthma symptoms and exacerbations include airway irritants (e.g., air pollutants, cold, heat, weather changes, strong odors or perfumes, smoke).

Most people who have asthma are sensitive to a variety of triggers.

ASTHMA PATHOPHYSIOLOGY



The underlying pathology in asthma is reversible and diffuse airway inflammation that leads to airway narrowing.

When activated, mast cells release several chemicals called mediators. These chemicals, which include histamine, prostaglandins, and leukotrienes, perpetuate the inflammatory response

causing increased blood flow, vasoconstriction, and, attraction of white blood cells to the area, and bronchoconstriction.

Clinical Manifestation



Cough



Dyspnea



Wheezing

An asthma exacerbation

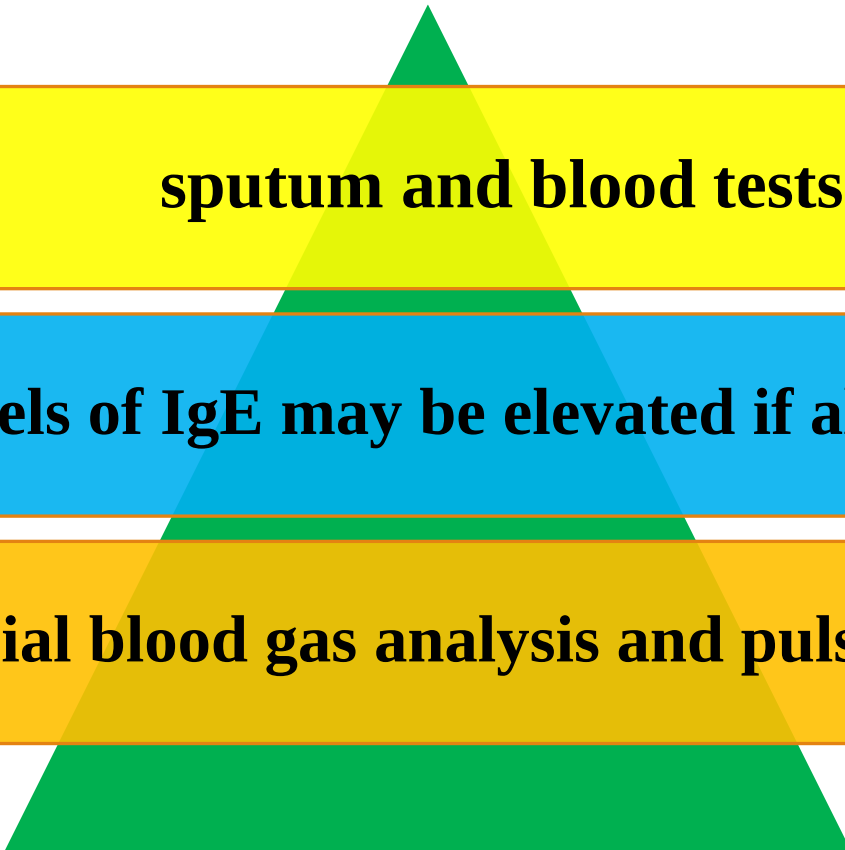
*May begin abruptly
but most frequently
is preceded by
increasing
symptoms over the
previous few days.*

There is cough, with
or without mucus
production.

As the exacerbation
progresses,
diaphoresis,
tachycardia, and a
widened pulse
pressure may occur
along with central
cyanosis

Assessment and Diagnostic Findings

- The clinician must determine that episodic symptoms of airflow obstruction are present.
- A positive family history and environmental factors, including climate changes (particularly cold air), and air pollution, are primarily associated with asthma.
- Comorbid conditions that may accompany asthma include gastro-esophageal reflux disease (GERD), and drug-induced asthma.



sputum and blood tests

Serum levels of IgE may be elevated if allergy is present

Arterial blood gas analysis and pulse oximetry

PREVENTION



Patients with recurrent asthma should undergo tests to identify the substances that precipitate the symptoms.



Dust



Dust mites



Roach



Pets



Horses



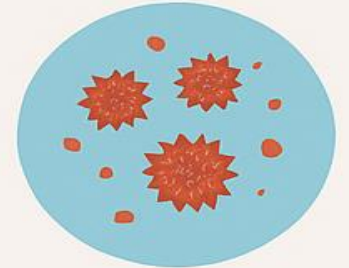
Detergents



Certain foods



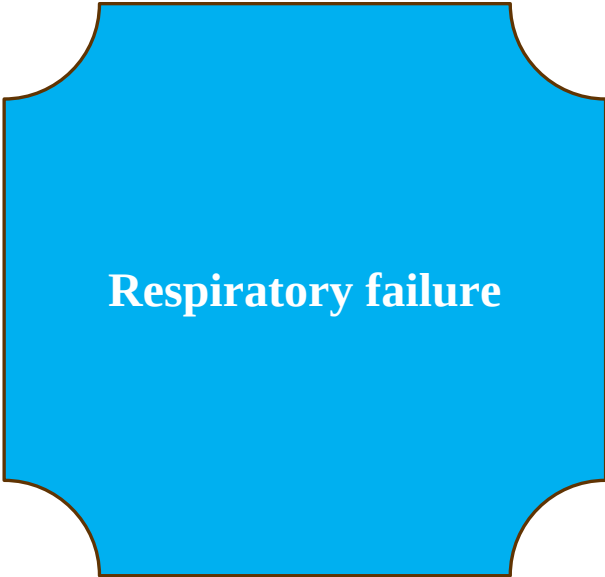
Molds



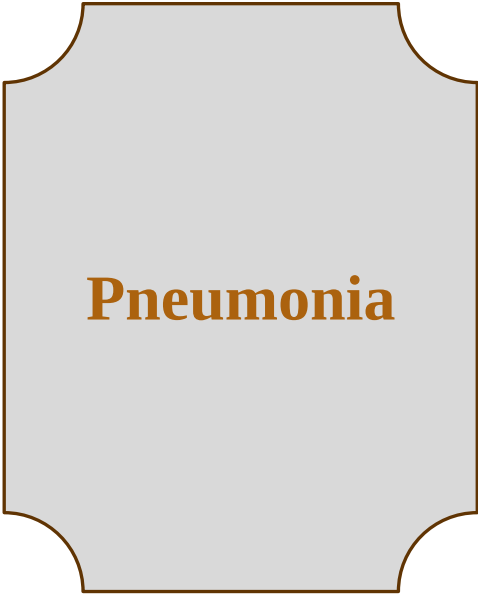
Pollen

Patients are instructed to avoid the causative agents whenever possible. Knowledge is the key to quality asthma care.

Complications



Respiratory failure



Pneumonia



Atelectasis

Medical Management



Immediate intervention may be necessary

I. Pharmacologic Therapy

II. Management of Exacerbations

III. Peak flow monitoring

Pharmacologic Therapy



1. Quick Relief Medications

- Short acting: - beta2 adrenergic agonists (Ventolin) are the medications of choice for relief of acute symptoms.
- Anticholinergics (e.g. Atrovent) may be used in patients who do not tolerate short acting beta2 adrenergic agonists.



2. Long-Acting Control Medications

- Corticosteroids are the most effective anti-inflammatory medications currently available. Initially, an inhaled form is used. Patients should rinse their mouth after administration to prevent thrush, a common complication associated with use of inhaled corticosteroids.



Ø Long acting beta2 adrenergic agonists are used with anti-inflammatory medications to control asthma symptoms, particularly those that occur during the night.

← ○ Salmeterol (Serevent) has duration of bronchodilator for at least 12 hours. It used with other medications in long term control of asthma.

CONTROL ASTHMA EXACERBATIONS

Avoid Triggers



Identify and avoid known triggers such as allergens, smoke, cold air, and medications. triggers

Adhere to Medications



Take prescribed medications including inhalers and controllers, as directed by your healthcare provider

Monitor Symptoms



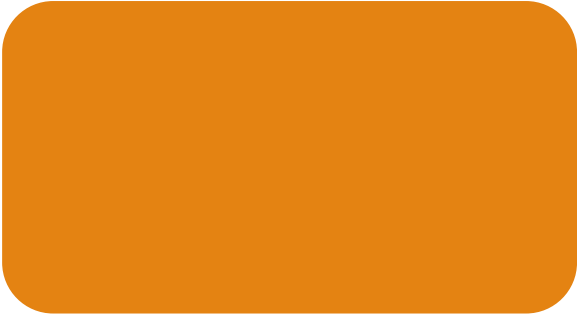
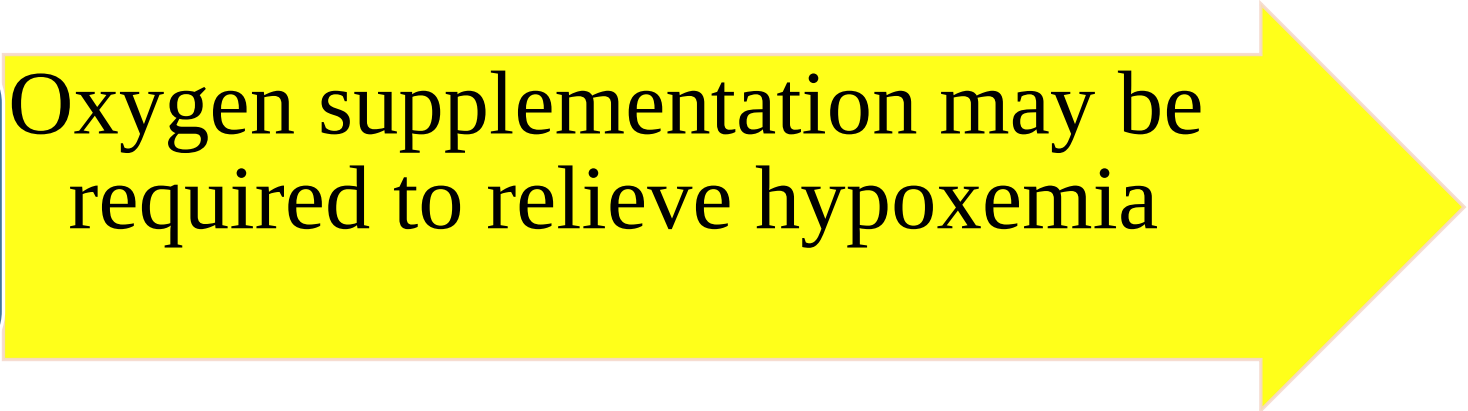
Regularly assess breathing and symptoms, use a flow meter if recommended

Management of Exacerbations

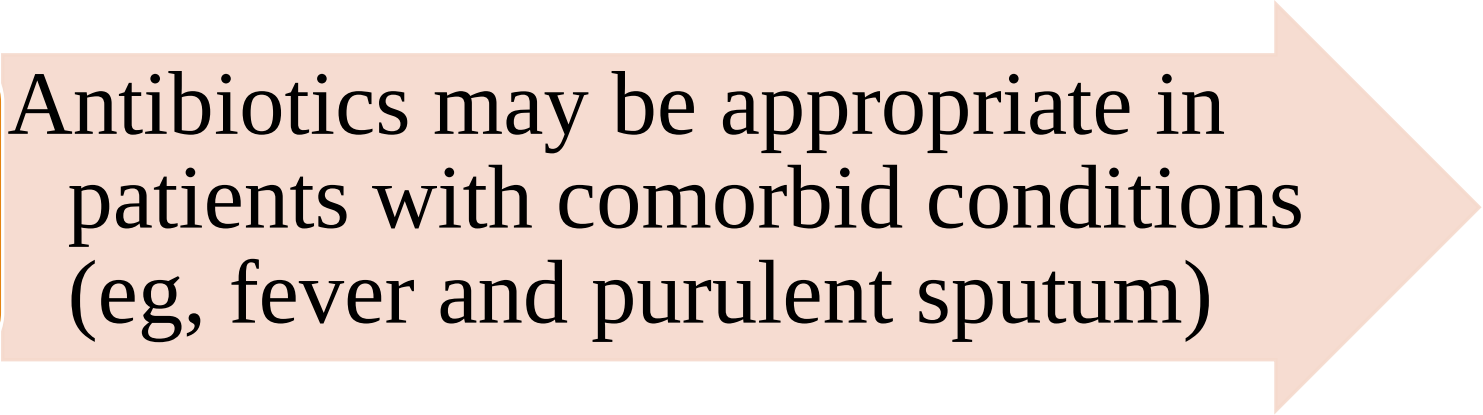
- ✓ Asthma exacerbations are best managed by early treatment and education, including education patients about self-management techniques, especially those with moderate or severe persistent asthma or with a history of severe exacerbations.



Oxygen supplementation may be required to relieve hypoxemia



Antibiotics may be appropriate in patients with comorbid conditions (eg, fever and purulent sputum)





Peak Flow Monitoring

1- Daily peak flow monitoring is recommended for patients who meet one or more of the following **criteria**:

- Have moderate or severe persistent asthma
- Worsening symptoms

2- The patient is instructed in the proper technique particularly about using maximal effort

Nursing Management

The immediate nursing care of patients with asthma depends on the severity of symptoms.

The patient may be treated successfully as an outpatient if asthma symptoms are relatively mild or may require hospitalization and intensive care if symptoms are acute and severe

The patient and family are often frightened and anxious because of the patient's dyspnea. Therefore, a calm approach is an important aspect of care.



The nurse assesses the patient's respiratory status by monitoring the severity of symptoms, breath sounds, pulse oximetry, and vital signs.

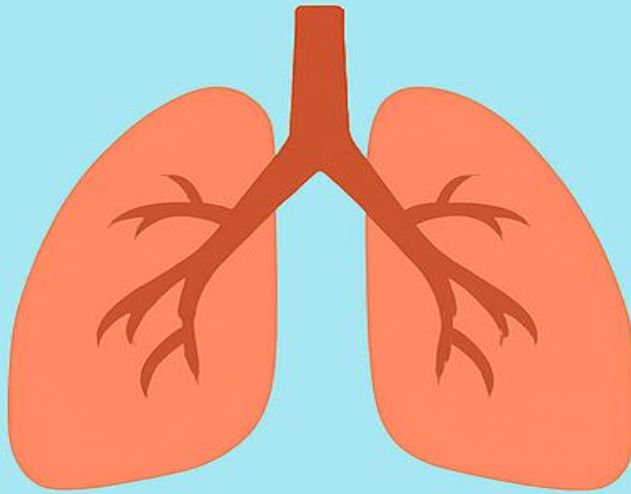


Obtains a history of allergic reactions to medications before administering medications

❑ Administers fluids if the patient is dehydrated

❑ If the patient requires intubation, the nurse assists with the intubation procedure, continues close monitoring of the patient

THANK YOU



ASTHMA